1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________________ May 8, 2016 Sunday

Department/School:____________________ SID:___________________________________

University Physics

Midterm Examination

Kuang Yaming Honors School, Nanjing University

Select five out of following six problems.

1.(20pts.) A diver of mass m jumps off a diving platform of height h above surface of the water with zero initial velocity. Assume that when she is in the water, the buoyant force of the water just balances the gravity and the viscous resistance of the water on the diver is bv^2, where v is the velocity of the diver and b is a constant.

(a) What is the initial velocity of the diver v_0 when she begins to dive into the water?

(b) Write down the equation of motion of the diver when she descends through the water.

(c) Find the depth at which the velocity v of the diver is half of the v_0.

(d) Find the vertical depth $x(t)$ of the diver in terms of the time under water.

2.(20pts.)A particle of mass m is subject to a one dimensional force $F(x) = -kx + bx^3$ where k and b are both positive constants.

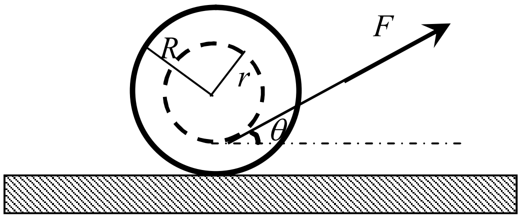
(a) Find the corresponding potential energy function of the particle, taking $x = 0$ as the reference point.

(b) Sketch the potential energy diagram. Show the kinetic energy of the particle for a given total mechanical energy.

(c) Determine the equilibrium points of the particle, and identify them as stable or unstable.

3. (20pts.) At time $t=0$, a satellite of mass m is at a distance of two radii of the earth ($2R_e$) from the center of the earth passing through the equatorial plane with a velocity v perpendicular to the plane. Suppose that $v=\alpha v_0$, $v_0=\sqrt{GM_e/2R_e}$, where $M_e >> m$ is the mass of the earth and α is a numerical factor. We treat the earth as though it were a perfect sphere.
(a) Find the value of α with which the orbit of the satellite is a circle.
(b) Find the range of α within which the satellite can escape the gravity of the earth.
(c) Find the range of α within which the satellite will hit the earth.

4. (20pts.) A spool of mass m and radius R rests on a rough horizontal surface. The moment of inertia of the spool to its axis is $I = \gamma m R^2$, where γ is a numerical factor. The spool is wound with a massless thread and the radius of the wound thread is $r \le R$. A constant force F that makes an angle θ to the horizontal is exerted on the thread so that the spool rolls without slipping on the surface, as shown in the figure.



(a) What is the translational acceleration of the spool? What is the rotational acceleration of the spool?

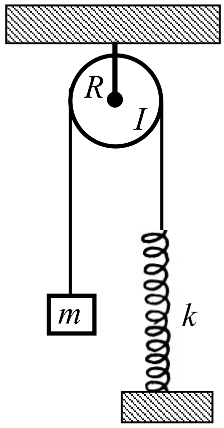
(b) What is the work done by the F during the first interval of time t after the beginning of the motion of the spool?

5. (20 pts.) A thin uniform square plate of side length l and mass M can rotate freely about a fixed vertical axis that is coinciding with one of its sides. The plate is at rest initially. A small solid sphere of mass m elastically strikes on the center of the plate with a velocity v that is perpendicular to the plate.

(a) Find the velocity of the sphere v' and the rotational velocity of the plate after the collision.

(b) Find the horizontal component of the force the axis exerts on the plate after the collision.

6. (20pts.) As shown in the accompanying figure, a pulley of radius R is hanging on the ceiling. The moment of inertia of it is I . A block of mass m is hanged on the left side of the string that is wrapped around the pulley. The other side of the string is attached to a spring of spring constant k that is fixed on the ground. The string is massless and it does not slip on the pulley. All other frictions are negligible.



(a) What is the equilibrium condition for the system?

(b) Find the angular frequency of the small oscillation of the arrangement.